

Sheet 05 — MA-INF 2201 — Computer Vision

1 Multi-Label MRFs

1.1 Alpha Expansion Graph Construction

Legend:

$i \in \{a, b, c, d, e\}$ node index

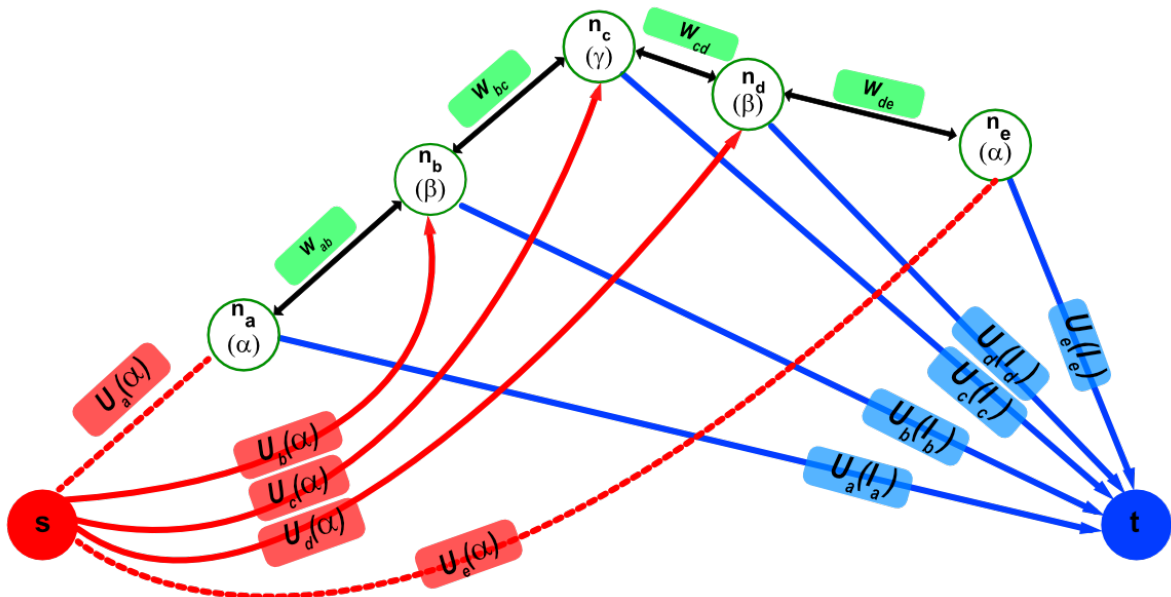
n_i node i in the graph

$s \rightarrow n_i : U_i(\alpha)$ switch node i to label α

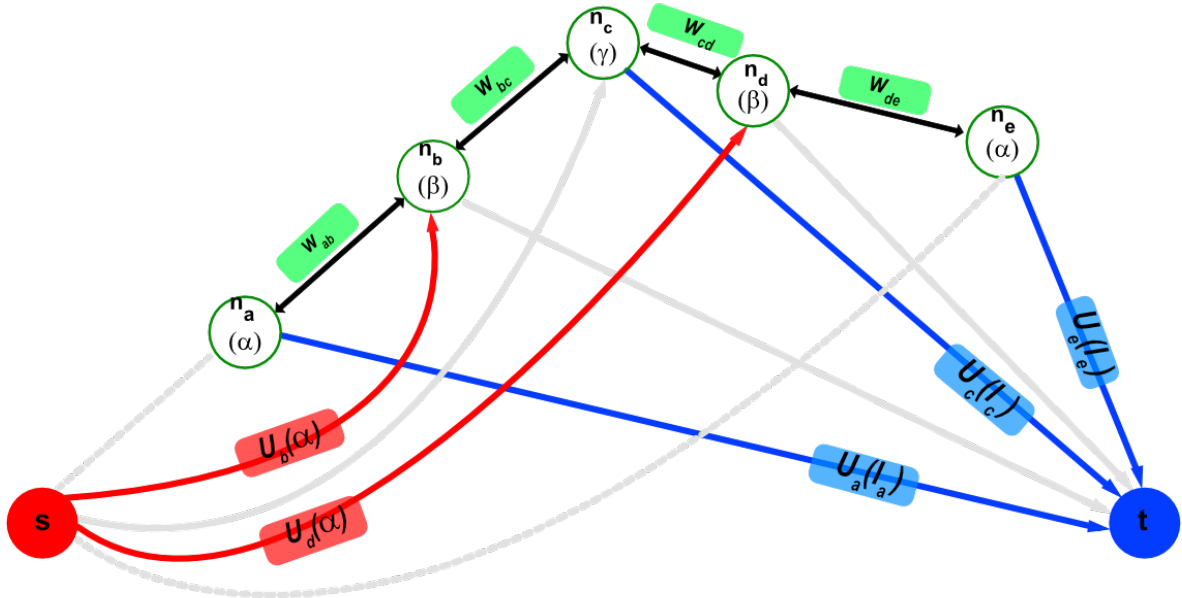
$n_i \rightarrow t : U_i(l_i)$ keep current label l_i of node i

$(n_i, n_j) : w_{ij}$ α -expansion reduced pairwise cost

$$w_{ij} = P_{ij}(l_i, l_j) + P_{ij}(\alpha, \alpha) - P_{ij}(l_i, \alpha) - P_{ij}(\alpha, l_j)$$



1.2 The Minimum Cut



Total Cost of This Configuration

$$C = U_a(l_a) + U_c(l_c) + U_e(l_e) + U_b(\alpha) + U_d(\alpha) + w_{ab} + w_{bc} + w_{cd} + w_{de}$$